MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

END OF SEMESTEREXAMINATION

ICT 4701: SOFTWARE ENGINEERING

TIME ALLOWED: 3 HOURS **DATE : FEBRUARY, 2018**

INSTRUCTIONS

i. This paper contains **two**pages. Please check.
ii. There are **four questions** in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

Question one

a. What are the **ten** important attributes that all professional software should have? **(10 marks)**

b. Describe the different consumers of software requirements i.e. the different users, or types of users, of a software requirements document. For each consumer, explain how he or she would use the requirements. **(10 marks)**

c. Describe the differences between unit and integration testing. Give the goals for each type of testing and describe when and how each should occur. **(6 marks)**

d. Explain why interface testing is necessary even when individual components have been extensively validated through component testing and program inspections. **(6 marks)**

e. Discuss risk management process. **(4 marks)**

Question two

a. Explain in detail why a software system that is used in a real-world environment must change or become progressively less useful. **(4 marks)**

b. Extreme programming express user requirements as stories, with each story written on a card. List and explain **two** advantages and **two** disadvantages of this approach to requirements description. **(4 marks)**

c. In computer security terms, explain the differences between an attack and a threat. **(4 marks)**

d. Discuss the following Software process models:
   i. Waterfall **(3 marks)**
   ii. Evolutionary **(3 marks)**

e. Explain the role of Joint Application Development (JAD) in Software Requirements. **(6 marks)**

Question three

a. Explain why the intangibility of software systems poses special problems for software project management. **(4 marks)**

b. Distinguish between user requirements and system requirements. **(4 marks)**

c. Discuss **two** problems which might be associated with use of natural language during requirements engineering/analysis. **(4 marks)**

d. List any **two** reasons why the productivity rate of programmers working as a pair is roughly the same as two programmers working individually. **(2 marks)**

e. Discuss the following software maintenance types.

I. Corrective. **(2 marks)**
II. Adaptive. **(2 marks)**
III. Perfective. **(2 marks)**
IV. Preventive. **(2 marks)**

f. Discuss the following software costing models:

I. COCOMO I. **(4 marks)**
II. COCOMO II. **(4 marks)**

g. Using your knowledge of how an ATM is used, develop a set of use-cases that could serve as a basis for understanding the requirements for an ATM system. **(10 marks)**

END OF QUESTION PAPER